

Supporting Information for Fast Generation of Pipek–Mezey Wannier Functions via the Co-Iterative Augmented Hessian Method

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I. STRUCTURE FILES

The structure files for all solid-state systems studied in this work are available in the following GitHub repository:

https://github.com/hongzhouye/supporting_data/tree/main/2026/PMWF

where

- the structure of h-BN, diamond, MgO, silicon, SiO₂, *trans*-(C₂H₂)_∞, C-nanotube, and graphene is taken from [J. Phys. Chem. A](#) **128**, 8570 (2024),
- the structure of CO/MgO(001) is taken from [Faraday Discuss.](#) **254**, 628 (2024),
- and the structure of aluminum (*fcc*) is generated with a lattice constant of 4.05 Å.

II. SUPPLEMENTARY TABLES

TABLE S1. CPU times (s) of key steps in k -point and Γ -point PMWF optimization for all systems considered in this work. **Proj.**: atomic projector initialization (for the meta-Löwdin projectors). **Obj.**: PM objective function evaluation. **Grad.**: gradient evaluation. **Hvp (init)**: precomputation for the Hessian–vector product. **Hvp**: Hessian–vector product evaluation. Note that atomic projector initialization is performed only once for the entire optimization, whereas the precomputation for the Hessian–vector product is performed once per CIAH iteration. The reported CPU times are averaged over 5 independent runs using the same computational resources described in Section III of the main text.

Main conclusion: (i) for the k -point based formulation, each Hessian–vector product evaluation is computationally less expensive than that of objective function and gradient evaluation, and (ii) all operations in the k -point based formulation are more efficient than their counterparts in the Γ -point formulation.

System	k -point					Γ -point				
	Proj.	Obj.	Grad	Hvp (init)	Hvp	Proj.	Obj.	Grad	Hvp (init)	Hvp
h-BN	29.22	0.48	0.79	0.79	0.31	154.21	2.19	5.23	6.17	3.93
Diamond	24.10	1.30	1.91	1.95	0.69	456.73	6.42	16.65	19.37	12.75
MgO	43.84	0.96	1.39	1.41	0.43	749.51	8.03	17.97	20.61	12.62
Silicon	23.30	0.77	1.21	1.23	0.41	500.65	6.82	19.72	22.89	15.93
SiO ₂	4.24	0.18	0.33	0.38	0.17	28.19	0.47	1.77	1.75	1.25
CO/MgO (100)	3.54	0.18	0.41	0.47	0.28	12.88	0.30	1.00	1.36	1.01
<i>trans</i> -(C ₂ H ₂) _∞	12.93	0.30	0.44	0.44	0.20	43.82	0.54	2.28	2.70	2.12
C-nanotube	7.10	0.46	0.95	1.03	0.53	79.86	1.28	3.01	3.36	2.54
Graphene	14.89	0.41	0.72	0.76	0.29	149.41	2.56	7.99	7.15	5.52
Aluminum	9.27	0.53	0.73	0.79	0.34	160.62	2.72	7.60	9.14	6.50

TABLE S2. Number of iterations required for k -BFGS to converge as a function of the maximum step size s_0 , defined in Eqn 38 of the main text. In most cases, the converged PMWFs are the same as those reported in Table 3 of the main text. Two exceptions are the C-nanotube for $s_0 \geq 0.30$ a.u. and aluminum for $s_0 = 0.20$ a.u., where local minima with a higher PM objective value are obtained. To balance efficiency and numerical stability, we adopt $s_0 = 0.10$ a.u. as the default.

System	Maximum step size, s_0 (a.u.)					
	0.05	0.10	0.20	0.30	0.50	1.00
h-BN	38	27	22	19	20	20
Diamond	70	51	34	27	21	119
MgO	32	24	23	22	22	22
Silicon	65	49	32	25	314	160
SiO ₂	374	312	425	350	259	303
CO/MgO (100)	287	187	204	212	200	208
<i>trans</i> -(C ₂ H ₂) _∞	39	27	35	19	31	17
C-nanotube	93	65	55	59*	50*	49*
Graphene	139	129	123	118	118	117
Aluminum	336	197	198*	181	182	240

TABLE S3. Same as Table 3 in the main text, except that the breakdown of PM objective function, gradient, and Hessian–vector product evaluations is shown.

System	k -CIAH			k -BFGS			Γ -CIAH		
	PM obj.	N_{iter}	$N_{\text{f}} + N_{\text{g}} + N_{\text{Hv}}$	PM obj.	N_{iter}	$N_{\text{f}} + N_{\text{g}}$	PM obj.	N_{iter}	$N_{\text{f}} + N_{\text{g}} + N_{\text{Hv}}$
h-BN	2.276	3	3 + 11 + 42	2.276	27	53 + 27	2.276	6	6 + 14 + 50
Diamond	1.859	4	4 + 15 + 62	1.859	51	101 + 51	1.859	5	5 + 15 + 57
MgO	3.241	9	9 + 24 + 105	3.241	19 + 5	46 + 24	3.241	9	9 + 26 + 80
Silicon	1.798	5	5 + 18 + 66	1.798	49	97 + 49	1.798	5	5 + 16 + 60
SiO ₂	17.657	15	15 + 40 + 184	17.657	306	621 + 306	17.657	7	7 + 21 + 74
CO/MgO (100)	61.650	19	19 + 68 + 325	61.650	187	378 + 187	61.650	9	9 + 34 + 137
<i>trans</i> -(C ₂ H ₂) _∞	3.343	4	4 + 13 + 45	3.343	27	53 + 27	3.343	4	4 + 11 + 35
C-nanotube	28.141	7	7 + 26 + 117	28.141	65	129 + 65	28.142	8	8 + 30 + 116
Graphene	2.801	5	5 + 22 + 96	2.801	129	257 + 129	2.801	9	9 + 30 + 133
Aluminum	2.683	12	12 + 52 + 249	2.684	197	395 + 197	2.683	11	11 + 37 + 154

TABLE S4. Same as Table 3 in the main text, except that intrinsic atomic orbitals are used as the atomic projectors. The relative convergence behavior of the three algorithms considered, k -CIAH, k -BFGS, and Γ -CIAH, closely follows the pattern observed in Table 3 of the main text, indicating that PMWF optimization is relatively insensitive to the choice of local projectors (here, IAOs versus meta-Löwdin projectors).

System	k -CIAH			k -BFGS			Γ -CIAH		
	PM obj.	N_{iter}	$N_{\text{f}} + N_{\text{g}} + N_{\text{Hv}}$	PM obj.	N_{iter}	$N_{\text{f}} + N_{\text{g}}$	PM obj.	N_{iter}	$N_{\text{f}} + N_{\text{g}} + N_{\text{Hv}}$
h-BN	2.130	4	4 + 18 + 67	2.130	27	54 + 27	2.130	5	5 + 14 + 52
Diamond	1.910	5	5 + 23 + 89	1.910	67	133 + 67	1.910	5	5 + 14 + 56
MgO	2.680	6	6 + 30 + 118	2.680	99	205 + 99	2.680	6	6 + 20 + 70
Silicon	1.876	4	4 + 20 + 88	1.876	41	81 + 41	1.876	6	6 + 16 + 56
SiO ₂	17.849	15	15 + 48 + 229	17.849	251	517 + 251	17.849	8	8 + 24 + 85
CO/MgO (100)	57.368	13	13 + 42 + 214	57.368	359	735 + 359	57.365	8	8 + 28 + 110
<i>trans</i> -(C ₂ H ₂) _∞	3.578	4	4 + 17 + 66	3.578	28	56 + 28	3.578	4	4 + 10 + 33
C-nanotube	28.576	7	7 + 32 + 142	28.575	66	132 + 66	28.576	7	7 + 23 + 90
Graphene	2.907	5	5 + 27 + 127	2.907	123	248 + 123	2.907	15	15 + 46 + 189
Aluminum	2.185	14	14 + 64 + 296	2.190	197	394 + 197	2.190	23	23 + 64 + 266

TABLE S5. Same as Table 3 in the main text, except that the convergence behavior of Γ -CIAH is compared using atomic initial guesses with and without phase alignment (PA). The “with PA” results are identical to those reported in Table 3. The “without PA” results are obtained from the atomic initial guess generated directly in the supercell, which is the default initial guess when the molecular CIAH code is applied directly to a periodic supercell with Γ -point Brillouin-zone sampling. The data suggest that the phase-aligned initial guess stabilizes the Γ -CIAH optimization by helping avoid convergence to local minima or saddle points (e.g., h-BN, diamond, and MgO). In the aluminum case, phase alignment enables successful convergence of an otherwise failed optimization after 1000 cycles.

System	Γ -CIAH (with PA)			Γ -CIAH (without PA)		
	PM obj.	N_{iter}	$N_{\text{f}} + N_{\text{g}} + N_{\text{Hv}}$	PM obj.	N_{iter}	$N_{\text{f}} + N_{\text{g}} + N_{\text{Hv}}$
h-BN	2.276	6	6 + 14 + 50	2.276	4 + 3	7 + 14 + 38
Diamond	1.859	5	5 + 15 + 57	1.859	10 + 7	17 + 36 + 140
MgO	3.241	9	9 + 26 + 80	3.241	4 + 3	7 + 15 + 45
Silicon	1.798	5	5 + 16 + 60	1.798	7	7 + 17 + 60
SiO ₂	17.657	7	7 + 21 + 74	17.657	7	7 + 20 + 73
CO/MgO (100)	61.650	9	9 + 34 + 137	61.650	7 + 3	10 + 27 + 92
<i>trans</i> -(C ₂ H ₂) _∞	3.343	4	4 + 11 + 35	3.343	5	5 + 11 + 29
C-nanotube	28.142	8	8 + 30 + 116	28.163	7	7 + 21 + 88
Graphene	2.801	9	9 + 30 + 133	2.800	22	22 + 72 + 342
Aluminum	2.683	11	11 + 37 + 154	Failed		

III. SUPPLEMENTARY FIGURES

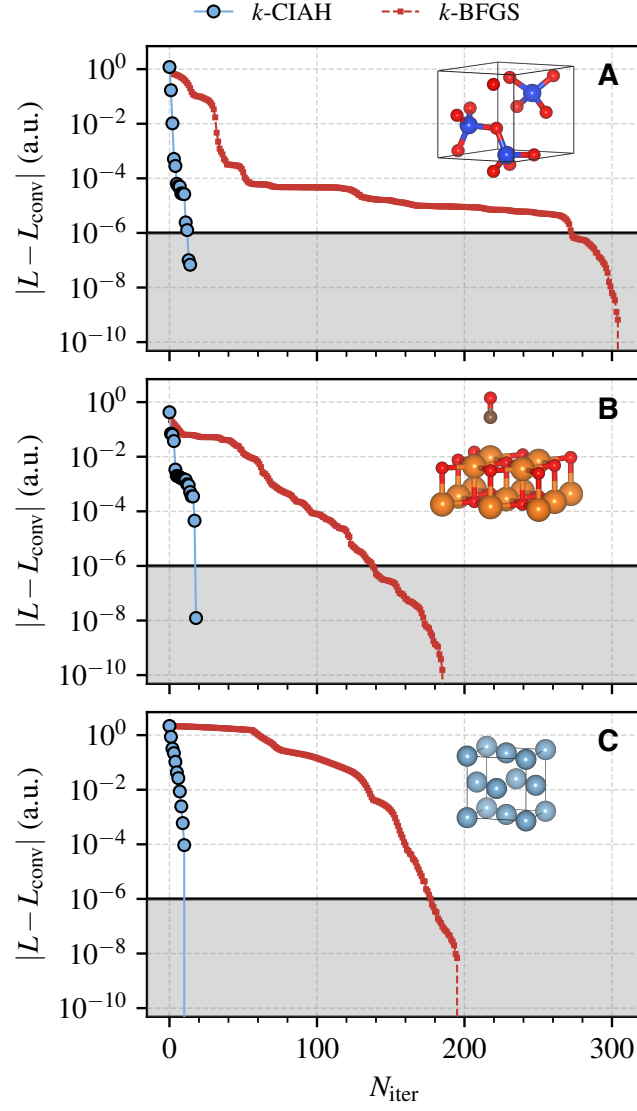


FIG. 1. Convergence of k -CIAH and k -BFGS measured by the decay of the absolute error of PM objective function L for (A) SiO₂, (B) CO/MgO(001), and (C) aluminum. The convergence threshold (10^{-6}) is denoted by the black horizontal line.

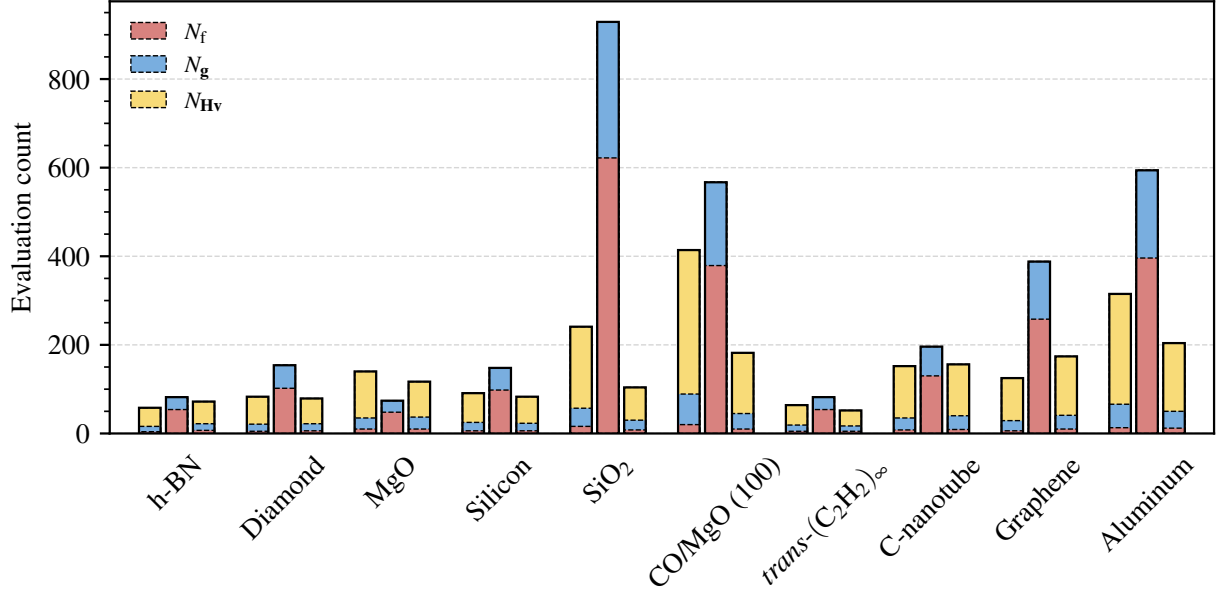


FIG. 2. Breakdown of the evaluation count of PM objective (N_f), gradient (N_g), and Hessian–vector product (N_{Hv}) for the ten systems discussed in Table 3 in the main text. The total count, i.e., the sum of the three numbers, for each system is the number reported in Table 3 in the main text.

IV. DERIVATION OF ANALYTICAL GRADIENTS, HESSIAN-VECTOR PRODUCT, AND HESSIAN DIAGONALS

A. Parameterization

As discussed in the main text, the k -space unitaries can be parameterized using k -dependent generators, each of which is a complex anti-Hermitian matrix. Let $\kappa_{\mathbf{k}} = X_{\mathbf{k}} + i Y_{\mathbf{k}}$ where both $X_{\mathbf{k}}$ and $Y_{\mathbf{k}} \in \mathbb{R}^{n_{\text{orb}} \times n_{\text{orb}}}$. The unique parameters are the lower-triangular part of $X_{\mathbf{k}}$, excluding the diagonals, and the lower-triangular part of $Y_{\mathbf{k}}$, including the diagonals.

B. Basic derivatives

Define $(E_{mn})_{ij} = \delta_{mi}\delta_{nj}$. The first- and second-order derivatives of the unitary with respect to the generators evaluated at $\kappa_{\mathbf{k}} = 0$ are

$$\begin{aligned} \frac{\partial U_{\mathbf{k}'}}{\partial X_{\mathbf{k}mn}} &= \delta_{\mathbf{k}\mathbf{k}'}(E_{mn} - E_{nm}), & (m \neq n) \\ \frac{\partial U_{\mathbf{k}'}}{\partial Y_{\mathbf{k}mn}} &= i\delta_{\mathbf{k}\mathbf{k}'}(E_{mn} + E_{nm}), & (m \neq n) \\ \frac{\partial U_{\mathbf{k}'}}{\partial Y_{\mathbf{k}mm}} &= i\delta_{\mathbf{k}\mathbf{k}'}E_{mm}. \end{aligned} \quad (1)$$

$$\frac{\partial^2 U_{\mathbf{k}''}}{\partial A_{\mathbf{k}mn} \partial B_{\mathbf{k}'pq}} = \frac{1}{2} \left\{ \frac{\partial U_{\mathbf{k}''}}{\partial A_{\mathbf{k}mn}}, \frac{\partial U_{\mathbf{k}''}}{\partial B_{\mathbf{k}'mn}} \right\}, \quad (A, B = X, Y) \quad (2)$$

C. Cost function

The PM objective function defined in the main text here is recapped here. A minus sign is added to the objective function to turn the maximization a minimization problem.

$$L = - \sum_{T\mathbf{A},i} Q_{T\mathbf{A},0i}^p \quad (3)$$

$$Q_{T\mathbf{A},0i} = P_{T\mathbf{A},0i,0i} = \sum_{\mathbf{k}\mathbf{k}'} \left(U_{\mathbf{k}}^\dagger P_{T\mathbf{A},\mathbf{k}\mathbf{k}'} U_{\mathbf{k}'} \right)_{ii} \quad (4)$$

$$P_{T\mathbf{A},\mathbf{k}\mathbf{k}'} = \frac{1}{N_k} \sum_{R\mathbf{R}'} \theta_{R\mathbf{k}}^* P_{T\mathbf{A},R\mathbf{R}'} \theta_{R'\mathbf{k}'} \quad (5)$$

D. Gradient

Consider the first-order variation of L ,

$$\delta L = -p \sum_{T_{A,i}} Q_{T_{A,0i}}^{p-1} \delta Q_{T_{A,0i}} \quad (6)$$

which depends on the first-order variation of Q ,

$$\delta Q_{T_{A,0i}} = 2\Re \sum_{k'} \left(P_{T_{A,0k'}} \delta U_{k'} \right)_{ii}, \quad \left(P_{T_{A,0k'}} := \sum_k P_{T_{A,kk'}} \right) \quad (7)$$

For the real and imaginary part of the generator, the derivatives are

$$\begin{aligned} \frac{\partial Q_{T_{A,0i}}}{\partial X_{kmn}} &= -2 \left(P_{T_{A,0m,kn}}^{\Re} \delta_{mi} - P_{T_{A,0n,km}}^{\Re} \delta_{ni} \right) \\ \frac{\partial Q_{T_{A,0i}}}{\partial Y_{kmn}} &= -2 \left(P_{T_{A,0m,kn}}^{\Im} \delta_{mi} + P_{T_{A,0n,km}}^{\Im} \delta_{ni} \right) \end{aligned} \quad (8)$$

The two can be combined conveniently into a complex-valued expression,

$$\frac{\partial Q_{T_{A,0i}}}{\partial X_{kmn}} + i \frac{\partial Q_{T_{A,0i}}}{\partial Y_{kmn}} = -2 \left(P_{T_{A,0m,kn}} \delta_{mi} - P_{T_{A,0n,km}}^* \delta_{ni} \right) \quad (9)$$

which leads to the following expression for the gradient,

$$\begin{aligned} G_k &= \tilde{G}_k - \tilde{G}_k^\dagger, \\ \tilde{G}_{kmn} &= 2p \sum_{T_A} Q_{T_{A,0m}}^{p-1} P_{T_{A,0m,kn}} \end{aligned}$$

(10)

Using $P_{T_{A,k0}} = P_{T_{A,0k}}^\dagger$, one can convert eq. (10) to Eqn 25 in the main text. We prefer to put k index first as in $P_{T_{A,k0}}$ due to the C-ordering of matrices and tensors in our code.

E. Hessian–vector product

Let v_k denote the vector to be applied by the Hessian. Let v_k^X and v_k^Y denote its real and imaginary parts, respectively. The Hessian–vector product is defined as follows.

$$\begin{aligned} \Sigma_{kmn}^X &= \frac{1}{2} \sum_{k'pq} \frac{\partial^2 L}{\partial X_{kmn} \partial X_{k'pq}} v_{k'pq}^X + \frac{\partial^2 L}{\partial X_{kmn} \partial Y_{k'pq}} v_{k'pq}^Y, \\ \Sigma_{kmn}^Y &= \frac{1}{2} \sum_{k'pq} \frac{\partial^2 L}{\partial Y_{kmn} \partial X_{k'pq}} v_{k'pq}^X + \frac{\partial^2 L}{\partial Y_{kmn} \partial Y_{k'pq}} v_{k'pq}^Y, \end{aligned} \quad (11)$$

where at a high-level, we need to evaluate the second-order variation of L ,

$$\delta^2 L = -p(p-1) \sum_{TAi} Q_{TA,0i}^{p-2} (\delta Q_{TA,0i})^2 - p \sum_{TAi} Q_{TA,0i}^{p-1} \delta^2 Q_{TA,0i} \quad (12)$$

The first term only depends on the first-order variation of Q , whose expression has already been derived above. We call this term *disconnected*. The second term depends on the second-order variation of Q ,

$$\delta^2 Q_{TA,0i} = 2\Re \sum_{\mathbf{k}'} \left(P_{TA,0\mathbf{k}'} \delta^2 U_{\mathbf{k}'} \right)_{ii} + 2\Re \sum_{\mathbf{k}\mathbf{k}'} \left(\delta U_{\mathbf{k}}^\dagger P_{TA,\mathbf{k}\mathbf{k}'} \delta U_{\mathbf{k}'} \right)_{ii} \quad (13)$$

and therefore is *connected*. We further differentiate the first and second term in $\delta^2 Q$ to be *asymmetric* and *symmetric* connected terms.

1. Disconnected term

$$\begin{aligned} \Sigma_{kmn}^X &= -\frac{1}{2}p(p-1) \sum_{TAi} Q_{TA,0i}^{p-2} \frac{\partial Q_{TA,0i}}{\partial X_{kmn}} \sum_{\mathbf{k}'pq} \left(\frac{\partial Q_{TA,0i}}{\partial X_{\mathbf{k}'pq}} v_{\mathbf{k}'pq}^X + \frac{\partial Q_{TA,0i}}{\partial Y_{\mathbf{k}'pq}} v_{\mathbf{k}'pq}^Y \right), \\ \Sigma_{mn}^Y &= -\frac{1}{2}p(p-1) \sum_{TAi} Q_{TA,0i}^{p-2} \frac{\partial Q_{TA,0i}}{\partial Y_{kmn}} \sum_{\mathbf{k}'pq} \left(\frac{\partial Q_{TA,0i}}{\partial X_{\mathbf{k}'pq}} v_{\mathbf{k}'pq}^X + \frac{\partial Q_{TA,0i}}{\partial Y_{\mathbf{k}'pq}} v_{\mathbf{k}'pq}^Y \right) \end{aligned} \quad (14)$$

where the term in the parenthesis can be simplified to

$$\sum_{\mathbf{k}'pq} \left(\frac{\partial Q_{TA,0i}}{\partial X_{\mathbf{k}'pq}} v_{\mathbf{k}'pq}^X + \frac{\partial Q_{TA,0i}}{\partial Y_{\mathbf{k}'pq}} v_{\mathbf{k}'pq}^Y \right) = 4\Re \sum_{\mathbf{k}'} \left(P_{TA,0\mathbf{k}'} v_{\mathbf{k}'} \right)_{ii} \quad (15)$$

Combining this with the gradient of Q [eq. (8)] gives

$$\begin{aligned} \Sigma_{\mathbf{k}} &= \tilde{\Sigma}_{\mathbf{k}} - \tilde{\Sigma}_{\mathbf{k}}, \\ \tilde{\Sigma}_{kmn} &= 4p(p-1) \sum_{TA} Q_{TA,0m}^{p-2} \sum_{\mathbf{k}'} \Re(P_{TA,0\mathbf{k}'} v_{\mathbf{k}'}')_{mm} P_{TA,0m,\mathbf{k}n} \end{aligned}$$

(16)

2. Connected symmetric term

$$\begin{aligned} \Sigma_{kmn}^X &= -p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re \left[\frac{\partial U_{\mathbf{k}}^\dagger}{\partial X_{kmn}} P_{TA,\mathbf{k}\mathbf{k}'} \sum_{\mathbf{k}'pq} \left(\frac{\partial U_{\mathbf{k}'}}{\partial X_{\mathbf{k}'pq}} v_{\mathbf{k}'pq}^X + \frac{\partial U_{\mathbf{k}'}}{\partial Y_{\mathbf{k}'pq}} v_{\mathbf{k}'pq}^Y \right) \right]_{ii}, \\ \Sigma_{kmn}^Y &= -p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re \left[\frac{\partial U_{\mathbf{k}}^\dagger}{\partial Y_{kmn}} P_{TA,\mathbf{k}\mathbf{k}'} \sum_{\mathbf{k}'pq} \left(\frac{\partial U_{\mathbf{k}'}}{\partial X_{\mathbf{k}'pq}} v_{\mathbf{k}'pq}^X + \frac{\partial U_{\mathbf{k}'}}{\partial Y_{\mathbf{k}'pq}} v_{\mathbf{k}'pq}^Y \right) \right]_{ii}, \end{aligned} \quad (17)$$

where the term in the parenthesis can be simplified to

$$\sum_{\mathbf{k}'_{pq}} \frac{\partial U_{\mathbf{k}'}}{\partial X_{\mathbf{k}'_{pq}}} v_{\mathbf{k}'_{pq}}^X + \frac{\partial U_{\mathbf{k}'}}{\partial Y_{\mathbf{k}'_{pq}}} v_{\mathbf{k}'_{pq}}^Y = 2 \sum_{\mathbf{k}'_{pq}} E_{pq} v_{\mathbf{k}'_{pq}} \quad (18)$$

Combining this with the gradient of U [eq. (1)] gives

$$\boxed{\begin{aligned} \Sigma_{\mathbf{k}} &= \tilde{\Sigma}_{\mathbf{k}} - \tilde{\Sigma}_{\mathbf{k}}^\dagger, \\ \tilde{\Sigma}_{\mathbf{k}mn} &= -2p \sum_{TA} Q_{TA,0n}^{p-1} \sum_{\mathbf{k}'} \left(P_{TA,\mathbf{k}\mathbf{k}'} v_{\mathbf{k}'} \right)_{mn} \end{aligned}} \quad (19)$$

3. Connected asymmetric term

$$\begin{aligned} \Sigma_{\mathbf{k}mn}^X &= -\frac{1}{2}p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re \left[P_{TA,0\mathbf{k}} \left\{ \frac{\partial U_{\mathbf{k}}}{\partial X_{\mathbf{k}mn}}, \sum_{pq} \frac{\partial U_{\mathbf{k}}}{\partial X_{\mathbf{k}pq}} v_{\mathbf{k}pq}^X + \frac{\partial U_{\mathbf{k}}}{\partial Y_{\mathbf{k}pq}} v_{\mathbf{k}pq}^Y \right\} \right]_{ii} \\ \Sigma_{\mathbf{k}mn}^Y &= -\frac{1}{2}p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re \left[P_{TA,0\mathbf{k}} \left\{ \frac{\partial U_{\mathbf{k}}}{\partial Y_{\mathbf{k}mn}}, \sum_{pq} \frac{\partial U_{\mathbf{k}}}{\partial X_{\mathbf{k}pq}} v_{\mathbf{k}pq}^X + \frac{\partial U_{\mathbf{k}}}{\partial Y_{\mathbf{k}pq}} v_{\mathbf{k}pq}^Y \right\} \right]_{ii} \end{aligned} \quad (20)$$

Using eq. (18) to simplify terms in the parenthesis leads to

$$\begin{aligned} \Sigma_{\mathbf{k}mn}^X &= -p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re \left[P_{TA,0\mathbf{k}} \left\{ \frac{\partial U_{\mathbf{k}}}{\partial X_{\mathbf{k}mn}}, \sum_{pq} E_{pq} v_{\mathbf{k}pq} \right\} \right]_{ii} \\ \Sigma_{\mathbf{k}mn}^Y &= -p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re \left[P_{TA,0\mathbf{k}} \left\{ \frac{\partial U_{\mathbf{k}}}{\partial Y_{\mathbf{k}mn}}, \sum_{pq} E_{pq} v_{\mathbf{k}pq} \right\} \right]_{ii} \end{aligned} \quad (21)$$

Using eq. (1) to further simplify the expression leads to

$$\boxed{\begin{aligned} \Sigma_{\mathbf{k}} &= \tilde{\Sigma}_{\mathbf{k}} - \tilde{\Sigma}_{\mathbf{k}}^\dagger, \\ \tilde{\Sigma}_{\mathbf{k}mn} &= p \sum_i v_{kmi} \left[\sum_{TA} Q_{TA,0i}^{p-1} (P_{TA,0\mathbf{k}})_{in} \right] + p \sum_{TA} Q_{TA,0m}^{p-1} (P_{TA,0\mathbf{k}} v_{\mathbf{k}})_{mn} \end{aligned}} \quad (22)$$

Finally, using $P_{TA,\mathbf{k}0} = P_{TA,0\mathbf{k}}^\dagger$, one can convert eqs. (16), (19) and (22) to Eqn 27–29 in the main text.

F. Hessian diagonal

The Hessian diagonal is defined as:

$$\begin{aligned} D_{\mathbf{k}mn}^X &= \frac{\partial^2 L}{\partial X_{\mathbf{k}mn}^2} = \tilde{D}_{\mathbf{k}}^X + \tilde{D}_{\mathbf{k}}^{X\top} \\ D_{\mathbf{k}mn}^Y &= \frac{\partial^2 L}{\partial Y_{\mathbf{k}mn}^2} = \tilde{D}_{\mathbf{k}}^Y + \tilde{D}_{\mathbf{k}}^{Y\top} \end{aligned} \quad (23)$$

where the second equality explicitly shows its symmetric nature.

1. *Disconnected term*

$$\begin{aligned}
D_{kmn}^X &= -p(p-1) \sum_{TA,i} Q_{TA,0i}^{p-2} \left(\frac{\partial Q_{TA,0i}}{\partial X_{kmn}} \right)^2 \\
&= -4p(p-1) \sum_{TA,i} Q_{TA,0i}^{p-2} [\Re(P_{TA,0k})_{mn}]^2 \delta_{mi} + (m \leftrightarrow n) \\
&= -4p(p-1) \sum_{TA} Q_{TA,0m}^{p-2} [\Re(P_{TA,0k})_{mn}]^2 + (m \leftrightarrow n)
\end{aligned} \tag{24}$$

$$\begin{aligned}
D_{kmn}^Y &= -p(p-1) \sum_{TA,i} Q_{TA,0i}^{p-2} \left(\frac{\partial Q_{TA,0i}}{\partial Y_{kmn}} \right)^2 \\
&= -4p(p-1) \sum_{TA,i} Q_{TA,0i}^{p-2} [\Im(P_{TA,0k})_{mn}]^2 \delta_{mi} + (m \leftrightarrow n) \\
&= -4p(p-1) \sum_{TA} Q_{TA,0m}^{p-2} [\Im(P_{TA,0k})_{mn}]^2 + (m \leftrightarrow n)
\end{aligned} \tag{25}$$

$$\boxed{
\begin{aligned}
\tilde{D}_{mn}^X &= -4p(p-1) \sum_{TA} Q_{TA,0m}^{p-2} [(\Re P_{TA,0k})_{mn}]^2 \\
\tilde{D}_{mn}^Y &= -4p(p-1) \sum_{TA} Q_{TA,0m}^{p-2} [(\Im P_{TA,0k})_{mn}]^2
\end{aligned}
} \tag{26}$$

2. *Connected asymmetric term*

$$\begin{aligned}
D_{kmn}^X &= -2p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re \left(P_{TA,0k} \frac{\partial^2 U_k}{\partial X_{kmn}^2} \right)_{ii} \\
&= -2p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re [P_{TA,0k} (E_{mn} - E_{nm})^2]_{ii} \\
&= 2p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re (P_{TA,0k} E_{mm})_{ii} + (m \leftrightarrow n)
\end{aligned} \tag{27}$$

$$\begin{aligned}
D_{kmn}^Y &= -2p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re \left(P_{TA,0k} \frac{\partial^2 U_k}{\partial Y_{kmn}^2} \right)_{ii} \\
&= 2p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re [P_{TA,0k} (E_{mn} + E_{nm})^2]_{ii} \\
&= 2p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re (P_{TA,0k} E_{mm})_{ii} + (m \leftrightarrow n)
\end{aligned} \tag{28}$$

Therefore

$$\boxed{\tilde{D}_{kmn}^X = \tilde{D}_{kmn}^Y = 2p \sum_{TA} Q_{TA,0m}^{p-1} (\Re P_{TA,0k})_{mm}} \quad (29)$$

3. Connected symmetric term

$$\begin{aligned} D_{kmn}^X &= -2p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re \left(\frac{\partial U_{\mathbf{k}}^\dagger}{\partial X_{kmn}} P_{TA,\mathbf{k}\mathbf{k}} \frac{\partial U_{\mathbf{k}}}{\partial X_{kmn}} \right)_{ii} \\ &= 2p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re [(E_{mn} - E_{nm}) P_{TA,\mathbf{k}\mathbf{k}} (E_{mn} - E_{nm})]_{ii} \\ &= -2p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re (E_{mn} P_{TA,\mathbf{k}\mathbf{k}} E_{nm})_{ii} + (m \leftrightarrow n) \end{aligned} \quad (30)$$

$$\begin{aligned} D_{kmn}^Y &= -2p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re \left(\frac{\partial U_{\mathbf{k}}^\dagger}{\partial Y_{kmn}} P_{TA,\mathbf{k}\mathbf{k}} \frac{\partial U_{\mathbf{k}}}{\partial Y_{kmn}} \right)_{ii} \\ &= -2p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re [(E_{mn} + E_{nm}) P_{TA,\mathbf{k}\mathbf{k}} (E_{mn} + E_{nm})]_{ii} \\ &= -2p \sum_{TA,i} Q_{TA,0i}^{p-1} \Re (E_{mn} P_{TA,\mathbf{k}\mathbf{k}} E_{nm})_{ii} + (m \leftrightarrow n) \end{aligned} \quad (31)$$

Therefore

$$\boxed{\tilde{D}_{kmn}^X = \tilde{D}_{kmn}^Y = -2p \sum_{TA} Q_{TA,0m}^{p-1} (\Re P_{TA,\mathbf{k}\mathbf{k}})_{nn}} \quad (32)$$

V. MOLECULAR CIAH-BASED PM OPTIMIZATION

The CIAH-based PM optimization for molecules and periodic solids in the Γ -point supercell formulation, as previously implemented in PySCF,¹ is computationally dominated by the Hessian-vector product,

$$(\mathbf{H}\mathbf{v})_{ij} = f_{ij} \hat{A}_{ij} \{ \tilde{\sigma}_{ij}^d + \tilde{\sigma}_{ij}^{\text{c-symm}} + \tilde{\sigma}_{ij}^{\text{c-asymm}} \}, \quad (33)$$

where the disconnected term is

$$\tilde{\sigma}_{ij}^d = -4p(p-1) \sum_A^{N_{\text{atom}}} Q_{A,j}^{p-2} \Re (v^\dagger P_A)_{jj} P_{A,ij}, \quad (34)$$

the connected symmetric term is

$$\tilde{\sigma}_{ij}^{\text{c-symm}} = -2p \sum_A^{N_{\text{atom}}} Q_{A,j}^{p-1} (P_A v)_{ij}, \quad (35)$$

and the connected asymmetric term is

$$\tilde{\sigma}_{ij}^{\text{c-asymm}} = -p \sum_m^{N_{\text{orb}}} \left(\sum_A Q_{A,m}^{p-1} P_{A,im} \right) v_{jm}^* - p \sum_A^{N_{\text{atom}}} Q_{A,j}^{p-1} (v^\dagger P_A)_{ij}. \quad (36)$$

Here,

$$P_{A,ij} = \langle w_i | \hat{P}_A | w_j \rangle = \sum_{\mu \in A} O_{\mu i}^* O_{\mu j} \quad (37)$$

is the projection matrix, and $Q_{A,i} = P_{A,ii}$ is the atomic population associated with localized orbital i . In eq. (37), $O_{\mu i}$ denotes the overlap between the projectors and the localized orbitals, which can be evaluated from their AO coefficient matrices as

$$O_{\mu i} = \langle \chi_\mu | w_i \rangle = \sum_{\rho\tau}^{N_{\text{AO}}^2} D_{\rho\mu}^* S_{\rho\tau} C_{\tau i}. \quad (38)$$

When the molecular CIAH algorithm is applied to a periodic supercell with Γ -point sampling, all indices are understood as supercell indices. Accordingly, $N_{\text{atom}} = N_k n_{\text{atom}}$, $N_{\text{orb}} = N_k n_{\text{orb}}$, $N_{\text{proj}} = N_k n_{\text{proj}}$, and $N_{\text{AO}} = N_k n_{\text{AO}}$.

As written, the CPU cost of the Hessian–vector product is dominated by forming the projection–vector product,

$$(P_A v)_{ij} = \sum_m^{N_{\text{orb}}} P_{A,im} v_{mj}, \quad (39)$$

which scales as $O(N_{\text{atom}} N_{\text{orb}}^3) = O(N_k^4 n_{\text{atom}} n_{\text{orb}}^3) \sim O(N_k^4 n^4)$. In this work, we employ an optimization analogous to that described in Section III E of the main text to improve the computational efficiency of molecular CIAH. In particular, the quartic scaling of eq. (39) can be reduced to cubic by exploiting the factorized form of the projection matrix in eq. (37),

$$(P_A v)_{ij} = \sum_{\mu \in A} O_{\mu i}^* \left(\sum_m^{N_{\text{orb}}} O_{\mu m} v_{mj} \right), \quad (40)$$

which can be evaluated in two steps, each scaling as $O(N_{\text{proj}} N_{\text{orb}}^2) = O(N_k^3 n_{\text{proj}} n_{\text{orb}}^2) \sim O(N_k^3 n^3)$.

The memory cost is dominated by storage of the projection matrix $P_{A,ij}$, whose size scales as $O(N_{\text{atom}} N_{\text{orb}}^2) \sim O(N_k^3 n^3)$. In principle, this cost could be reduced to $O(N_k^2 n^2)$ by using eq. (37) directly whenever $P_{A,ij}$ is required. In practice, however, we retain the explicit $P_{A,ij}$ tensor, which substantially simplifies the evaluation of the gradient and the remaining contributions to the Hessian–vector product.

VI. JACOBI SWEEP

A. General consideration

A general real-valued, translationally symmetric Jacobi rotation can be written as

$$\begin{bmatrix} \tilde{w}_{\mathbf{R}_0,i} & \tilde{w}_{\mathbf{R}_0+\mathbf{R},j} \end{bmatrix} = \begin{bmatrix} w_{\mathbf{R}_0,i} & w_{\mathbf{R}_0+\mathbf{R},j} \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, \quad (41)$$

which mixes $w_{\mathbf{0},i}$ with $w_{\mathbf{R},j}$ and, by translational covariance, applies the same rotation to all their lattice translates. Fourier transforming eq. (41) yields the corresponding k -space formulation,

$$\begin{bmatrix} \tilde{\phi}_{\mathbf{k}i} & \tilde{\phi}_{\mathbf{k}j} \end{bmatrix} = \begin{bmatrix} \phi_{\mathbf{k}i} & \phi_{\mathbf{k}j} \end{bmatrix} \begin{bmatrix} \cos \theta & e^{i\mathbf{k}\cdot\mathbf{R}} \sin \theta \\ -e^{-i\mathbf{k}\cdot\mathbf{R}} \sin \theta & \cos \theta \end{bmatrix}, \quad (42)$$

which corresponds to a k -dependent 2×2 rotation between Bloch orbitals i and j .

B. Working equations for $p = 2$

To determine the optimal rotation angle θ , consider the PM objective for a pair of WFs $(\mathbf{0}i, \mathbf{R}j)$ after applying the Jacobi rotation in eq. (41):

$$\tilde{L}_{ij} = \sum_{\mathbf{T}\mathbf{A}} [(\tilde{P}_{\mathbf{T}\mathbf{A},\mathbf{00}})_{ii}^2 + (\tilde{P}_{\mathbf{T}\mathbf{A},\mathbf{00}})_{jj}^2]. \quad (43)$$

Using

$$\begin{aligned} \tilde{w}_{\mathbf{0}i} &= w_{\mathbf{0}i} \cos \theta - w_{\mathbf{R}j} \sin \theta, \\ \tilde{w}_{\mathbf{0}j} &= w_{-\mathbf{R}i} \sin \theta + w_{\mathbf{0}j} \cos \theta, \end{aligned} \quad (44)$$

the diagonal projection matrix elements after rotation become

$$\begin{aligned} (\tilde{P}_{\mathbf{T}\mathbf{A},\mathbf{00}})_{ii} &= (P_{\mathbf{T}\mathbf{A},\mathbf{00}})_{ii} \cos^2 \theta + (P_{\mathbf{T}\mathbf{A},\mathbf{R}\mathbf{R}})_{jj} \sin^2 \theta - 2\Re(P_{\mathbf{T}\mathbf{A},\mathbf{0}\mathbf{R}})_{ij} \cos \theta \sin \theta, \\ (\tilde{P}_{\mathbf{T}\mathbf{A},\mathbf{00}})_{jj} &= (P_{\mathbf{T}\mathbf{A},(-\mathbf{R})(-\mathbf{R})})_{ii} \sin^2 \theta + (P_{\mathbf{T}\mathbf{A},\mathbf{00}})_{jj} \cos^2 \theta + 2\Re(P_{\mathbf{T}\mathbf{A},(-\mathbf{R})\mathbf{0}})_{ij} \cos \theta \sin \theta \\ &= (P_{\mathbf{T}\mathbf{A},\mathbf{00}})_{ii} \sin^2 \theta + (P_{\mathbf{T}\mathbf{A},\mathbf{R}\mathbf{R}})_{jj} \cos^2 \theta + 2\Re(P_{\mathbf{T}\mathbf{A},\mathbf{0}\mathbf{R}})_{ij} \cos \theta \sin \theta. \end{aligned} \quad (45)$$

In the second line for $(\tilde{P}_{\mathbf{T}\mathbf{A},\mathbf{00}})_{jj}$, we have used translational symmetry together with the fact that $\mathbf{T}$ is summed over in the final objective. The change in the PM objective due to the $(\mathbf{0}i, \mathbf{R}j)$ rotation is therefore

$$\Delta L_{ij}(\theta) = \tilde{L}_{ij}(\theta) - L_{ij} = -\frac{1}{2} \sin 2\theta [(A_{\mathbf{0}\mathbf{R}})_{ij} \cos 2\theta + (B_{\mathbf{0}\mathbf{R}})_{ij} \sin 2\theta], \quad (46)$$

where we have introduced the intermediates

$$\begin{aligned}(A_{\mathbf{0R}})_{ij} &= 4 \sum_{TA} [(P_{TA,\mathbf{00}})_{ii} - (P_{TA,\mathbf{RR}})_{jj}] \Re(P_{TA,\mathbf{0R}})_{ij}, \\ (B_{\mathbf{0R}})_{ij} &= \sum_{TA} [(P_{TA,\mathbf{00}})_{ii} - (P_{TA,\mathbf{RR}})_{jj}]^2 - [2 \Re(P_{TA,\mathbf{0R}})_{ij}]^2.\end{aligned}\tag{47}$$

Setting $\Delta L'_{ij}(\theta) = 0$ yields

$$\tan 4\theta = -\frac{(A_{\mathbf{0R}})_{ij}}{(B_{\mathbf{0R}})_{ij}}.\tag{48}$$

Equation (48) can be interpreted as follows:

1. Since the input orbitals are already at a stationary point of the PM functional, $\theta = 0$ is a trivial solution of eq. (48), implying that the right-hand side vanishes. The non-trivial solutions therefore satisfy $\tan 4\theta = 0$, i.e., $4\theta = n\pi$ with $n \neq 0$.
2. Furthermore, it is sufficient to restrict θ to the interval $[0, \pi)$, because a rotation by $\theta = \pi$ corresponds to a simultaneous sign change of the two orbitals, which leaves the PM objective invariant, and any $\theta > \pi$ can be mapped back into $[0, \pi)$.

We therefore conclude that the only distinct non-trivial solutions are

$$\theta = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}.\tag{49}$$

In practice, we can calculate ΔL_{ij} for all possible θ values in eq. (49) to find one that maximizes the PM objective.

C. Extension for $p > 2$

The derivation above strictly assumes $p = 2$. A similar derivation for $p = 3$ leads to the same condition eq. (48) for determining θ except that the two intermediates are defined as

$$\begin{aligned}(A_{\mathbf{0R}})_{ij} &= 4 \sum_{TA} [(P_{TA,\mathbf{0R}})_{ii}^2 - (P_{TA,\mathbf{0R}})_{jj}^2] \Re(P_{TA,\mathbf{0R}})_{ij}, \\ (B_{\mathbf{0R}})_{ij} &= \sum_{TA} [(P_{TA,\mathbf{0R}})_{ii} + (P_{TA,\mathbf{0R}})_{jj}] \left\{ [(P_{TA,\mathbf{0R}})_{ii} - (P_{TA,\mathbf{0R}})_{jj}]^2 - [2 \Re(P_{TA,\mathbf{0R}})_{ij}]^2 \right\},\end{aligned}\tag{50}$$

For general $p > 3$, an analytic derivation becomes cumbersome. However, the second argument above (periodicity in θ) always holds, and we have numerically validated that the first argument holds for $p \leq 6$ (which arguably already covers most use cases of PM localization). We therefore *postulate* that the first argument remains valid for all $p \geq 2$.

D. Efficient implementation and computational cost

In practice, we evaluate $\Delta L_{ij}(\theta)$ numerically for the three candidate angles in eq. (49) and check whether any of them increases the PM objective. The computational cost of this Jacobi stability check scales as

$$O(N_k^2 n_{\text{proj}} n_{\text{orb}}^2) \quad (51)$$

for building the $(P_{TA, \mathbf{k}\mathbf{R}})_{ij}$ intermediate. This cost is modest and comparable to that of a single gradient evaluation.

However, one can reasonably expect that pairwise instabilities primarily involve orbital pairs that are not too far apart in real space. We therefore can optionally restrict the real-space shifts in eq. (42) to satisfy

$$|\mathbf{R}| < R_{\text{max}}, \quad (52)$$

for a chosen cutoff R_{max} , which provides a controllable reduction in computational cost. Let N_R denote the number of lattice vectors $\mathbf{R}$ satisfying eq. (52). Under this restriction, the cost of the Jacobi stability check becomes

$$O(N_k N_R n_{\text{proj}} n_{\text{orb}}^2), \quad (53)$$

which is linear in N_k when N_R is independent of N_k . We choose $R_{\text{max}} = 10$ Bohr in this work, which covers WF pairs that are separated by roughly 5 chemical bonds.

REFERENCES

¹Q. Sun, “Co-iterative augmented hessian method for orbital optimization,” arXiv preprint arXiv:1610.08423 (2016), [arXiv:1610.08423](https://arxiv.org/abs/1610.08423) [physics.chem-ph].